INCH-POUND
MIL-DTL-1183/11B
w/AMENDMENT 1
20 August 2019
SUPERSEDING
MIL-DTL-1183/11B
20 December 2016

DETAIL SPECIFICATION SHEET

FITTINGS, PIPE, CAST BRONZE, SILVER-BRAZING; UNION ELBOW, PIPE AND UNION TEE, PIPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-1183.

REQUIREMENTS:

1. Female union elbow type. 45-degree and 90-degree union elbows with female tailpiece (see figures 1 and 2) shall be Type A (streamlined interior).

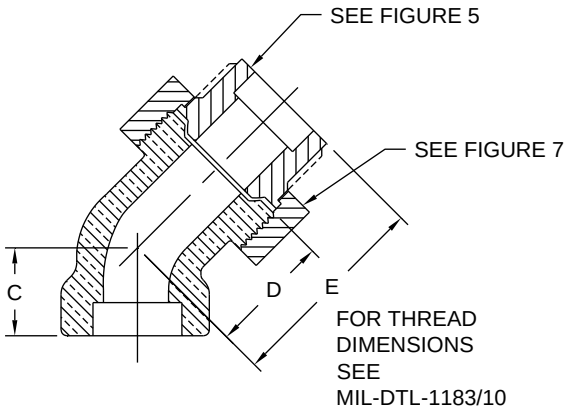


FIGURE 1. 45-degree union elbow.

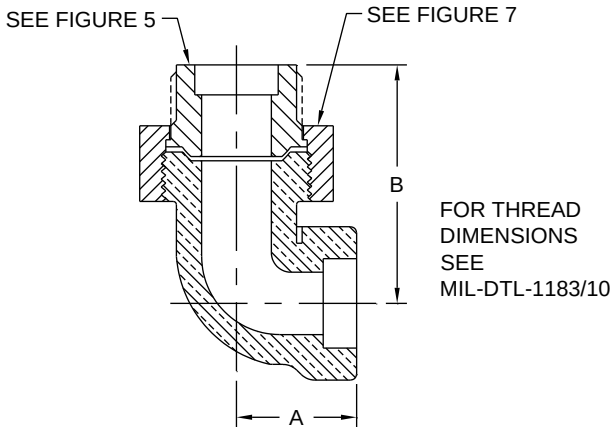


FIGURE 2. 90-degree union elbow.

2. Male union elbow type. 45-degree and 90-degree union elbows with male tailpiece shall be Type A (streamlined interior).

3. Female union tee type. Union branch and union run tees with male tailpiece (see figures 3 and 4) shall be Type A (streamlined interior).

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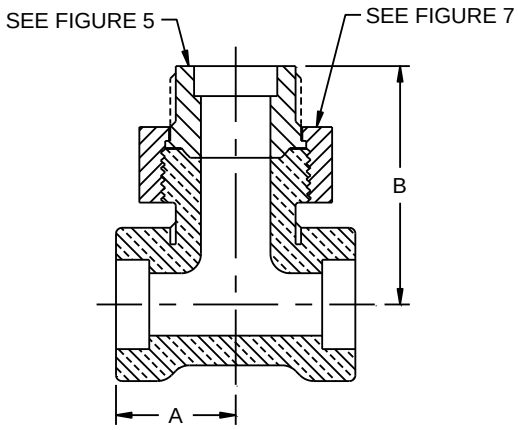


FIGURE 3. Union branch tee.

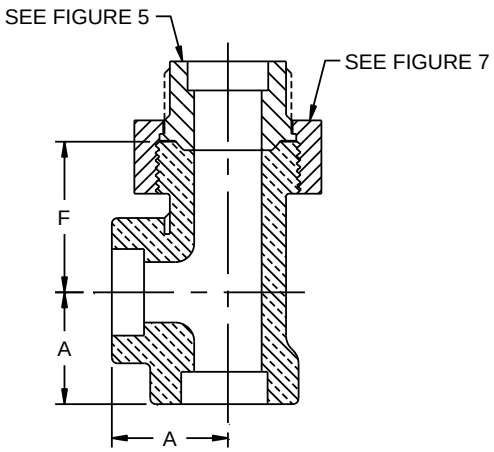


FIGURE 4. Union run tee.

4. Male union tee type. Union branch and union run tees with male tailpiece shall be Type A (streamlined interior).

5. Center-to-end dimensions. The center-to-end dimensions for union elbows and union tees shall be as specified in table I.

TABLE I. Center-to-end dimensions, inches.

Nominal pipe size	A	B	C	D	E	F
¼	15/16	1 15/16	13/16	25/32	1 9/16	1 5/32
⅜	1 1/16	2 7/32	⅞	55/64	1 ¾	1 11/32
½	1 ¼	2 13/32	1	1	1 15/16	1 15/32
¾	1 7/16	2 ⅞	1 1/8	1 1/8	2 3/16	1 13/16
1	1 5/8	3 5/32	1 5/16	1 19/64	2 ½	1 31/32
1 ¼	1 15/16	3 21/32	1 ½	1 ½	2 ⅞	2 19/64
1 ½	2 1/8	3 15/16	1 11/16	1 21/32	3 1/8	2 7/16
2	2 ½	4 ½	2	1 15/16	3 11/16	2 ¾

6. Female tailpiece dimensions. Dimensions for the female tailpiece (see figure 5) shall be as specified in table II.

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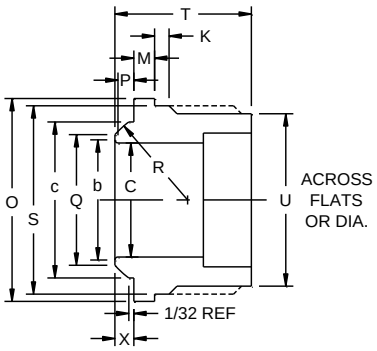


FIGURE 5. Female tailpiece.

7. Male tailpiece dimensions. Dimensions for the threaded male tailpiece (see figure 6) shall be as specified in table II. Male tailpiece dimensions are the same as the dimensions for the female tailpiece (see figure 5) except for those dimensions identified on figure 6. On figure 5, the dimensions P, Q, and R define the point of contact of the ground seat. NOTE: The size of the male pipe thread is one size larger than the nominal pipe size by which the union elbow or union tee is identified.

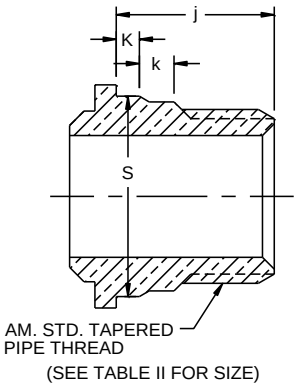


FIGURE 6. Male thread tailpiece.

TABLE II. Union components dimensions, inches.

Nominal pipe size	C Diameter		E	F ±0.016	G	H Min	I Diameter	K	L Min	M ±0.010
¼	0.398	±0.015	1.095	0.120	1 11/32	21/32	0.860	0.125	0.125	0.125
⅜	0.532	±0.015	1.281	0.125	1 35/64	45/64	1.020	0.135	0.135	0.135
½	0.697	±0.020	1.491	0.135	1 25/32	47/64	1.220	0.145	0.145	0.145
¾	0.907	±0.025	1.835	0.150	2 3/16	7/8	1.510	0.160	0.160	0.160
1	1.171	±0.025	2.165	0.170	2 35/64	59/64	1.820	0.175	0.175	0.180
1¼	1.502	±0.030	2.601	0.200	3 3/64	1 1/32	2.220	0.205	0.205	0.210
1½	1.742	±0.030	2.921	0.220	3 25/64	1 1/16	2.520	0.220	0.220	0.230
2	2.186	±0.030	3.521	0.250	4 3/64	1 5/32	3.090	0.250	0.250	0.260

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TABLE II. Union components dimensions, inches – Continued.

Nominal pipe size	O Diameter	P	Q	R Radius	S Diameter	T ±0.020	U	V	X	Pipe thread size
¼	0.965	0.100	0.510	0.361	0.83	0.874	0.75	16-UNS	0.134	⅜
⅜	1.135	0.103	0.650	0.460	0.99	0.959	0.90	14-UNS	0.139	½
½	1.345	0.113	0.835	0.591	1.19	1.049	1.09	14-UNS	0.159	¾
¾	1.650	0.115	1.030	0.728	1.47	1.154	1.35	11-UNS	0.154	1
1	1.975	0.135	1.315	0.930	1.78	1.299	1.63	11-UNS	0.184	1¼
1¼	2.395	0.130	1.650	1.167	2.18	1.469	2.00	10-UNS	0.174	1½
1½	2.710	0.143	1.915	1.355	2.47	1.619	2.27	10-UNS	0.199	2
2	3.310	0.165	2.395	1.694	3.04	1.854	2.80	10-UNS	0.224	2½

TABLE II. Union components dimensions, inches – Continued.

Nominal pipe size	b Diameter	c Diameter	f Pitch diameter		h Minor diameter		j	k
			Min	Max	Min	Max		
¼	0.441	0.618	1.0520	1.0585	1.0278	1.0348	0.985	0.30
⅜	0.576	0.768	1.2320	1.2390	1.2043	1.2122	1.155	0.30
½	0.741	0.972	1.4420	1.4492	1.4111	1.4190	1.165	0.30
¾	0.951	1.174	1.7720	1.7801	1.7326	1.7424	1.350	0.31
1	1.216	1.494	2.1020	2.1102	2.0626	2.0724	1.505	0.33
1¼	1.561	1.828	2.5321	2.5407	2.4887	2.4992	1.535	0.33
1½	1.801	2.116	2.8520	2.8609	2.8087	2.8192	1.610	0.39
2	2.276	2.638	3.4520	3.4610	3.4087	3.4192	2.020	0.39

8. Nut dimensions. Dimensions for the union nut (see figure 7) shall be as specified in table II.

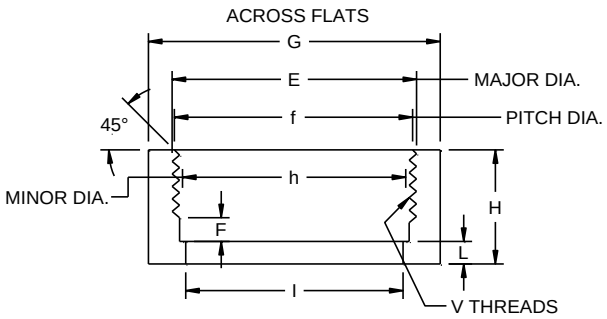


FIGURE 7. Nut.

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9. Static O-ring seal and retainer ring. When specified (see requirement 12 herein), union elbows and union tees shall be furnished with an O-ring and a retainer ring which, when installed, are outboard of the ground joint (see figure 8). The O-ring and retainer ring do not affect the primary contact of the ground seat and form an effective secondary seal. NOTE: For Navy applications, union elbows and union tees cannot be used without O-ring and retainer ring.

10. Retainer ring dimensions and material. Retainer ring dimensions shall be as specified in table III. The material of the retainer ring shall be ASTM B584 (UNS no. C90300) or ASTM B61 (UNS no. C92200) bronze (see figure 9). For prohibited materials, see paragraph 3.2 of the base document.

11. O-ring seal configuration and material. O-ring configuration shall be in accordance with SAE-AS568 of size specified in table IV for the required union elbow or union tee nominal pipe size. O-ring material shall be SAE-AMS-7276 Class 1 rubber and SAE-AS3208 and SAE-AS3209 for part number and sizing (see figure 10).

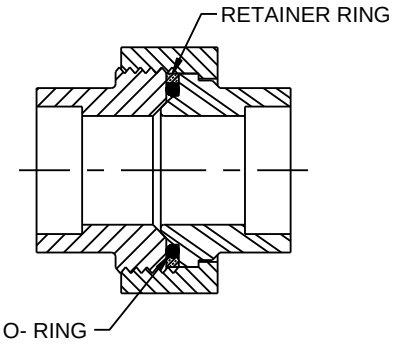


FIGURE 8. O-ring and retainer ring placement.

TABLE III. Retainer ring dimensions, inches.

Nominal pipe size	D ±.005	E ±.005	L ±.005
¼	0.050	0.813	1.008
⅜	0.050	0.938	1.184
½	0.050	1.188	1.394
¾	0.060	1.500	1.715
1	0.070	1.750	2.045
1¼	0.070	2.062	2.471
1½	0.070	2.375	2.791
2	0.090	3.000	3.391

TABLE IV. O-ring configuration and size.

Nominal pipe size (inches)	O-ring configuration and size
¼	AS568-017
⅜	AS568-019
½	AS568-023
¾	AS568-125
1	AS568-129
1¼	AS568-134
1½	AS568-139
2	AS568-232

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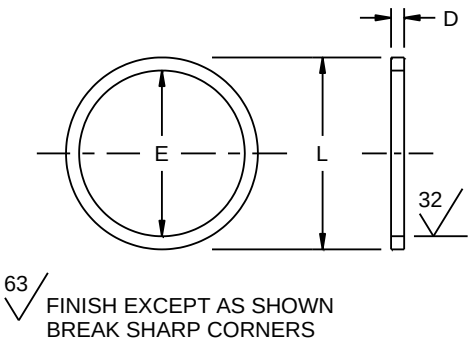


FIGURE 9. Retainer ring.

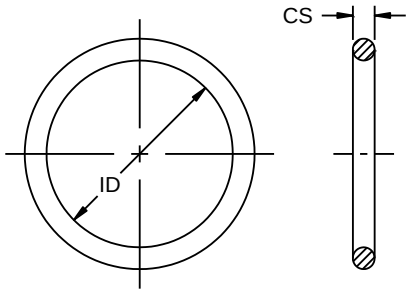


FIGURE 10. O-ring.

12. Specification sheet part number. The specification sheet part number is a definitive part number that is formed by combining the specification sheet code number (M1183/11 for MIL-DTL-1183/11) with a dash after it, the union-end fitting and pipe size code number, the brazing ring groove and code letter, the O-ring and retainer ring code number, and an optional dash and lead-free fitting identifier. Union elbows and union tees covered by this specification sheet shall be designated in the following form (see 6.2.b of MIL-DTL-1183):

M1183/11 – XX X X - PW

Specification sheet code number

Union-end fitting and pipe size code number (see requirement 13 herein)

Brazing ring groove and brazing ring code letter (see requirement 14 herein)

O-ring and retainer ring code number (see requirement 15 herein)

Lead-free fitting identifier (optional) (see 3.2 of MIL-DTL-1183)

13. Union-end fitting and pipe size. The union-end fitting (45-degree union elbow, 45-degree male union elbow, 90-degree union elbow, 90-degree male union elbow, union branch tee, male union branch tee, union run tee, or male union run tee) and the nominal pipe sizes are identified by two digits (see table V).

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TABLE V. Code number to union-end fitting and pipe size.

45-degree union elbow	45-degree male union elbow	90-degree union elbow	90-degree male union elbow	Union branch tee	Male union branch tee	Union run tee	Male union run tee	Nominal pipe size (inches)
11	21	31	41	51	61	71	81	¼
12	22	32	42	52	62	72	82	⅜
13	23	33	43	53	63	73	83	½
14	24	34	44	54	64	74	84	¾
15	25	35	45	55	65	75	85	1
16	26	36	46	56	66	76	86	1¼
17	27	37	47	57	67	77	87	1½
18	28	38	48	58	68	78	88	2

14. Brazing ring groove and brazing ring. Face fed (no groove), grooved without brazing ring, grooved with brazing ring inserted in groove, grooved with brazing rings separately packaged and attached to the union-end fitting, or grooved with brazing rings separately packaged and shipped with but not attached to the union-end fitting socket ends are identified by a single letter (see table VI).

TABLE VI. Code letter to socket condition.

Code letter	Condition of socket
N	No groove (face fed)
G 1/	Grooved (no brazing ring)
R 1/	Grooved with brazing ring inserted
A 2/	Grooved with brazing ring separately packaged and attached
S 1/ 2/	Grooved with brazing ring separately packaged and shipped with

NOTES:

1/ No National Stock Number (NSN) shall be established or exist in the Federal Supply System for a union-end fitting with the code letter of G, R, or S.

2/ When the socket condition code letter is A or S, the brazing rings shall not be inserted in the grooves, but shall be packaged separately. For condition code letter A, the packaged brazing ring shall be either included within each individual fitting package or physically attached to the fitting. For condition code letter S, the brazing ring shall be included within each individual fitting package.

15. O-ring seal and retainer ring. Whether the union-end fitting is provided without O-ring and retainer ring, with O-ring and retainer ring installed, with O-ring and retainer ring separately packaged and attached to the union-end fitting, or with O-ring and retainer ring separately packaged and shipped with the union-end fitting but not attached to the union-end fitting, it is identified by a digit (see table VII).

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TABLE VII. Code number to O-ring and retainer ring requirement.

Code number	O-ring and retainer ring requirement
1/ 2	Without O-ring and retainer ring
3	With O-ring (air, oil, or water service) and retainer ring installed
4	With O-ring (steam service) and retainer ring installed
1/ 5	With O-ring (air, oil, or water service) and retainer ring separately packaged and attached
1/ 6	With O-ring (steam service) and retainer ring separately packaged and attached
1/ 7	With O-ring (air, oil, and water service) and retainer ring separately packaged and shipped with
1/ 8	With O-ring (steam service) and retainer ring separately packaged and shipped with

NOTE:
1/ No National Stock Number (NSN) shall be established or exist in the Federal Supply System for a union-end fitting with a code number of 2, 5, 6, 7, or 8.

AMENDMENT NOTATIONS. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodians:
Army – CR4
Navy – SH

Preparing activity:
Navy – SH
(Project 4730-2019-031)

Review activity:
DLA – CC

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